



Question Set

Question 1

Consider a short option book hedged with the underlying and one liquid option. Define the state variables (spot, volatility, delta, gamma, vega, hedge ratios, and rebalance dates) and derive the local PnL expansion and hedge residual. State the measure, filtration, and tradability assumptions.

$$\Delta V \approx \Delta \Delta S + \frac{1}{2} \Gamma (\Delta S)^2$$

Question 2

For a short option book hedged with the underlying and one liquid option, explain which probability measure is natural for pricing. Write the density-process logic needed if the starting model is under the physical measure.

Question 3

Identify the discounted tradable or density-adjusted process in a short option book hedged with the underlying and one liquid option. Prove the martingale condition and name one modelling choice that would break it.

Question 4

Using delta -25, gamma 8, vega 42, spot shocks of plus/minus 0.4 percent, and a 2 vol-point move, construct a two-step numerical calculation. Show intermediate state values before discounting or aggregating.

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Question 5

For a short option book hedged with the underlying and one liquid option, compute and interpret delta-gamma-vega neutralization and residual basis risk. Explain the hedge sign and the unit of each sensitivity.

Question 6

Suppose the desk observes hedge instrument sensitivities observed from the risk system. Write the calibration objective, the weights you would use, and the parameter-identifiability checks.

Question 7

State the terminal and boundary conditions for a short option book hedged with the underlying and one liquid option. Explain how payoff, hedge unwind, and liquidity constraints prevents economically impossible prices.

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Question 8

Design a Monte Carlo or lattice estimator for a short option book hedged with the underlying and one liquid option. Specify daily delta-hedging error across stochastic spot paths, the estimator, the discounting rule, and the standard error.

Question 9

A junior quant implements a short option book hedged with the underlying and one liquid option and gets a stable but wrong answer. Explain how marking hedge PnL with stale Greeks after a large market move creates the error and design a diagnostic test.

Question 10

Write an NumPy algorithm for a short option book hedged with the underlying and one liquid option. Include inputs, outputs, diagnostics, and one benchmark case.

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Question 11

Starting from the payoff definition of a short option book hedged with the underlying and one liquid option, derive a replication, dynamic-programming, or conditional-expectation representation without skipping the economic argument.

Question 12

Create a stressed-market version of delta -25, gamma 8, vega 42, spot shocks of plus/minus 0.4 percent, and a 2 vol-point move. Recompute the value or risk metric and explain which term dominates the change.

Question 13

Use a second-order Taylor approximation for a short option book hedged with the underlying and one liquid option. Separate linear exposure, convexity exposure, cross terms if relevant, and residual model risk.

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Question 14

Show how a change of numeraire or change of measure would simplify one expectation in a short option book hedged with the underlying and one liquid option. Identify what becomes a martingale after the change.

Question 15

Give a proof-style answer showing when the wealth process for a short option book hedged with the underlying and one liquid option is self-financing. Explain why predictable trading strategies matter.

Question 16

Market quotes for hedge instrument sensitivities observed from the risk system are noisy. Propose a robust calibration workflow with initial guesses, constraints, residual plots, and out-of-sample checks.

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Question 17

For a short option book hedged with the underlying and one liquid option, compare the mathematical boundary condition with the trader's intuition. Give one limiting case where the answer should be obvious.

Question 18

Design a convergence study for daily delta-hedging error across stochastic spot paths. Specify path counts, time-step refinement, confidence intervals, and the exact quantity plotted.

Question 19

Find one way that marking hedge PnL with stale Greeks after a large market move could pass a superficial unit test. Write a stronger test that would fail for the biased implementation.

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Question 20

Explain a short option book hedged with the underlying and one liquid option and code the core routine. Give the derivation outline, pseudocode, and three numerical sanity checks.